Truly anonymous synthetic data - evolving legal definitions and technologies (Part I)

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The very first question commonly asked with respect to Synthetic Data is: **how accurate is it?** Seeing, then, the unparalleled accuracy of MOSTLY AI's synthetic data platform in action, resulting in synthetic data that is near indistinguishable from real data, typically triggers the second, just as important question: **is it really private?** In this blog post series we will dive into the topic of privacy preservation, both from a legal as well as a technical perspective, and explain how external assessors unequivocally come to the same conclusion: **MOSTLY AI generates truly anonymous synthetic data**, and thus any further processing, sharing or monetization becomes instantly compliant with even the toughest privacy regulations being put in place these days.

What is anonymous data? A historical perspective#

So, what does it mean for a dataset to be considered private in the first place? A common misconception is that any dataset that has its uniquely identifying attributes removed is thereby made anonymous. However, the absence of these so-called “direct identifiers”, like the full names, the e-mail addresses, the social security numbers, etc., within a dataset provides hardly any safety at all. The simple composition of multiple attributes taken together is unique and thus allows for the re-identification of individual subjects. Such attributes that are not direct identifiers by themselves, but only in combination, are thus termed quasi-identifiers. Just how easy this re-identification is, and how few attributes are actually required, came as a surprise to

a broader audience when Latanya Sweeney published her seminal paper entitled "Simple Demographics Often Identify People Uniquely" in the year 2000. In particular, because her research was preceded by the successful re-identification of publicly available highly sensitive medical records, that were supposedly anonymous. Notably, the simple fact that zip code, date-of-birth, and gender in combination are unique to 87% of the US population, served as an eye-opener for policy-makers and researchers alike. And as a direct result of Sweeney's groundbreaking work, privacy regulations were adapted to provide extra safety. For instance, the HIPAA privacy rule that covers the handling of health records in the US was passed in 2003, and explicitly lists a vast number of attributes (incl. any dates), that are all to be removed to provide safety (see footnote 15 here). Spurred by the discovery of these risks in methods then considered state-of-the-art anonymization, researchers continued their quest for improved de-identification methods and privacy guarantees, based on the concept of quasi-identifiers and sensitive attributes.

However, the approach ultimately had to hit a dead end, because **any** seemingly innocuous piece of **information**, that is related to an individual subject, is potentially sensitive, as well as **serves as a quasi-identifier**. And with more and more data points being collected, there is no escape from adversaries combining all of these with ease to single out individuals, even in large-scale datasets. The futility of this endeavor becomes even more striking once you realize that it's a hard **mathematical law** that one attempts to compete with: the so-called curse of dimensionality. The number of possible outcomes **grows exponentially** with the number of data points per subject, and even for modestly sized databases quickly surpasses the number of atoms in the universe. Thus, with a continuously growing number of data points collected, any individual in any database is increasingly distinctive from all others and therefore becomes susceptible to de-anonymization.

The consequence? Modern-day privacy regulations (GDPR, CCPA, etc.) provide **stricter definitions of anonymity**. They consider data to be anonymous if and only if none of the subjects are re-identifiable, neither directly nor indirectly, neither by the data controller nor by any third party (see GDPR §4.1). They do soften the definition, requiring that the re-identification attack needs to be reasonably likely to be performed (see GDPR Recital 26 or CCPA 1798.140(o)). However, these clauses clearly do NOT exempt from considering the most basic and thus reasonable form, the so-called linkage attacks, that simply require 3rd party data to be joined for a successful de-anonymization. And it was simple linkage attacks, that lead to the re-identification of Netflix users, the re-identification of NY taxi trips, the re-identification of telco location data, the re-identification of credit card transactions, the re-identification of browsing data, the re-identification of health care records, and so forth.

A 2019 Nature paper thus had to conclude:

> "[..] even heavily sampled anonymized datasets are unlikely to satisfy the modern standards for anonymization set forth by GDPR and seriously challenge the technical and legal adequacy of the de-identification release-and-forget model."

Gartner projects that by 2023, 65% of the world's population will have its personal information covered under modern privacy regulations, up from 10% today with the European GDPR regulation becoming the de-facto global standard.

True anonymization today is almost impossible with old tools#

Figure 1. The privacy-utility trade-off for existing de-identification techniques: generalization, masking, and obfuscation.

And while thousands, rather millions of bits are being collected about each one of us on a continuous basis, only 33 bits of information turn out to be sufficient to re-identify each individual among a global population of nearly 8 billion people. This gap between how much data is captured and how much information can be at most retained is widening by the day, making it an ever-more pressing concern for any organization dealing with privacy-sensitive data.

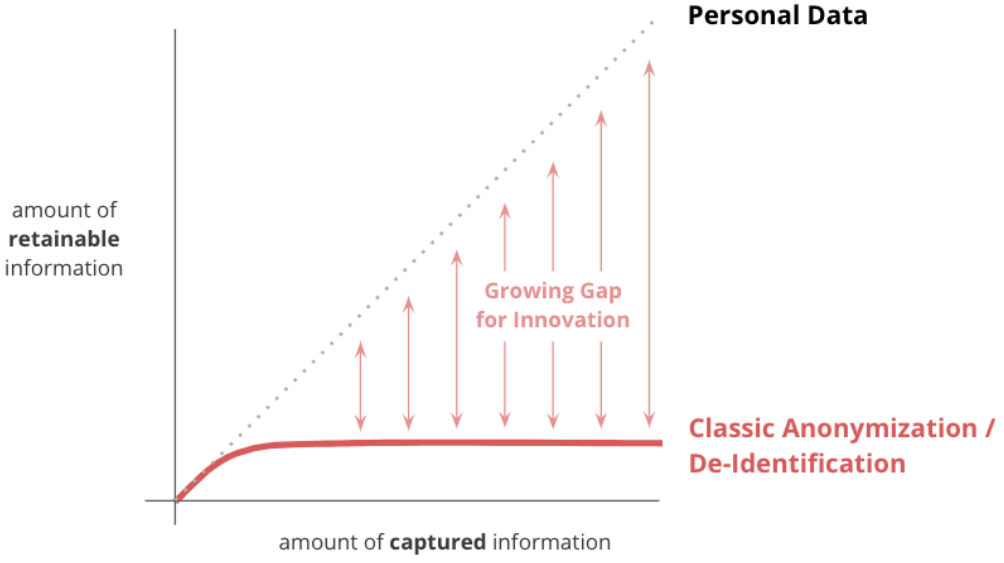

Figure 2. The increasing innovation gap between captured and retainable information.

In particular, as this limitation is true for any of the existing techniques, and for any real-world customer data asset, as has already been conjectured in the seminal 2008 paper on de-anonymizing basic Netflix ratings:

> "[..] the amount of perturbation that must be applied to the data to defeat our algorithm will **completely destroy their utility** [..] Sanitization techniques from the k-anonymity literature such as generalization and suppression do not provide meaningful privacy guarantees, and in any case fail on high-dimensional data."

So, true anonymization is hard. Is this the end of privacy? No! But we will need to move beyond mere de-identification. Enter synthetic data. Truly anonymous synthetic data.

Head over to Part II of our blog post series to dive deeper into the privacy of synthetic data, and how it can be assessed.

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